



IDBI 2026 INN@VATE

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Team Details

- a. **Team name:** [Team Name]
- b. **Team leader name:** [Team Leader Name]
- c. **Problem Statement:** Retail lending relies on bureau scores and static financials, causing low conversion, poor intent signals, inaccurate income estimation and high manual underwriting effort.

Brief about the idea

LendIQ — AI-Powered Retail Lending Intelligence Platform

LendIQ ingests consented Account Aggregator bank-statement data and turns it into pre-qualified, explainable loan leads for Personal, Home, Mortgage and Auto Loans — replacing static bureau-score checks with real cash-flow intelligence.

- **Parses** salary, EMI, rent, UPI, investments, cash deposits and 20 more transaction categories from raw AA JSON
- **Scores** income, repayment capacity, borrowing intent and default risk with ML models (LightGBM, XGBoost, SHAP)
- **Recommends** the right loan product, amount, rate and tenure — explained in plain language by a GenAI advisor

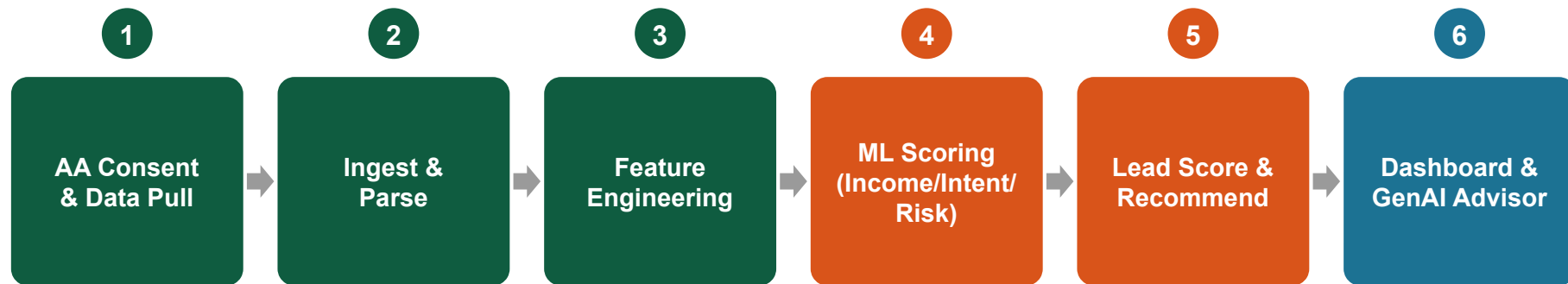
Opportunities

- **Differentiation** — Goes beyond bureau scores by mining actual Account Aggregator cash-flow behaviour: salary regularity, EMI load and real life-event spends, not a static credit snapshot.
- **Problem-solving** — Automates income estimation, repayment capacity, borrowing intent and risk that today need manual statement reading, cutting underwriting effort and turnaround time.
- **USP** — Explainable AI (SHAP) on every prediction plus a GenAI advisor grounded in the same scored profile — underwriters see the WHY, not a black box.

List of features offered by the solution

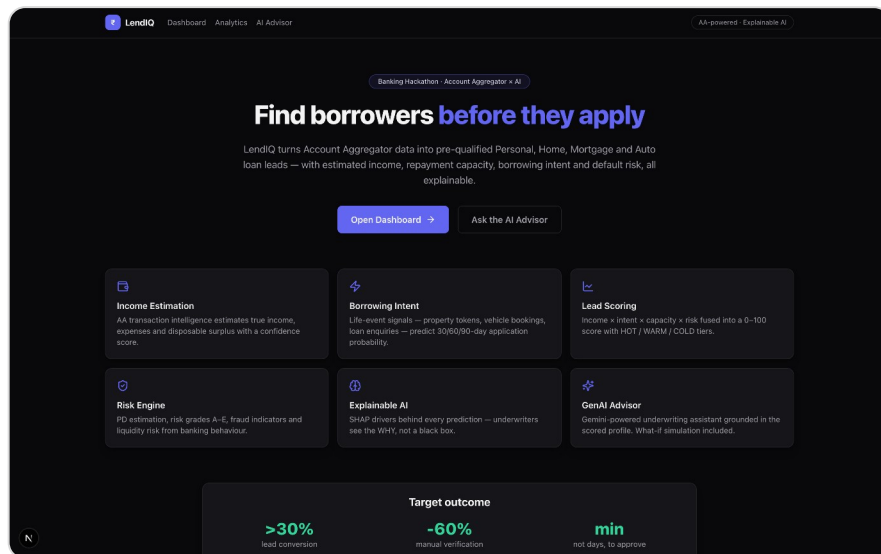
- **AA transaction parser** — 25-category classification from raw AA JSON
- **Income estimation** — confidence-scored monthly/net/disposable income
- **Repayment capacity** — FOIR, DTI, eligible EMI headroom
- **Borrowing intent** — 30/60/90-day application probability + reasons
- **Risk engine** — PD, A–E grade, fraud indicators, liquidity risk
- **AI lead scoring** — 0–100 score, HOT / WARM / COLD tiers
- **Loan recommendation** — 4 products, priced & explained offers
- **Explainable AI** — SHAP drivers behind every prediction
- **GenAI advisor** — Gemini chat + what-if repayment simulator
- **Executive dashboard** — KPIs, funnel, analytics & lead queue

Process flow diagram or Use-case diagram

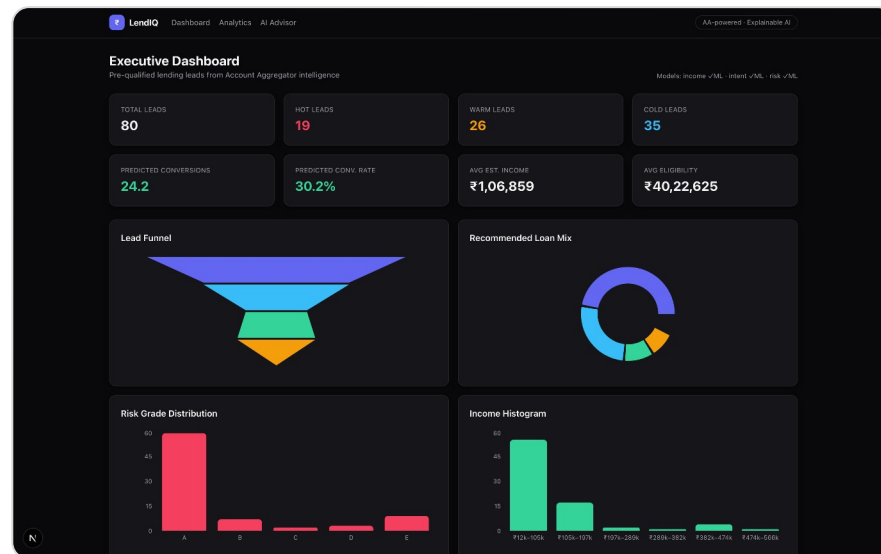


A customer's Account Aggregator consent triggers a one-way pipeline: raw bank transactions are parsed into behavioural features, scored by three ML engines in parallel, fused into a single lead score, and surfaced with an explainable recommendation — end to end in seconds.

Wireframes/Mock diagrams of the proposed solution (optional)

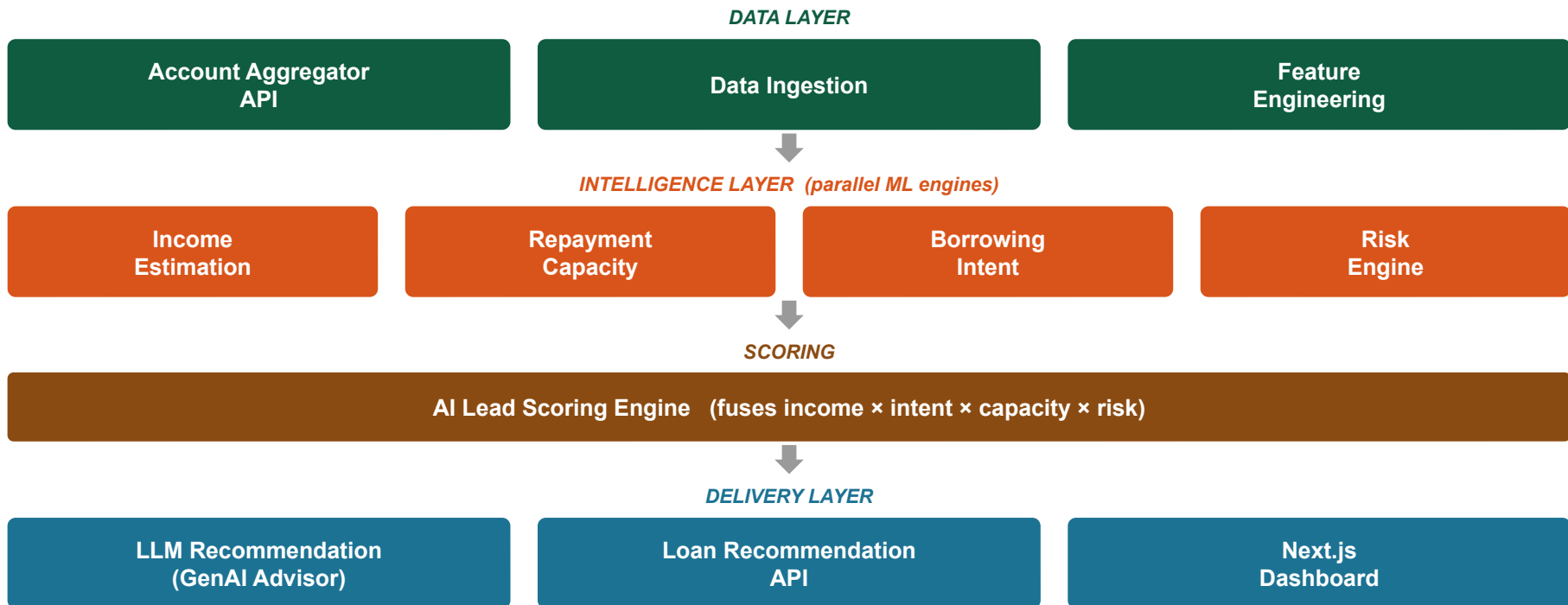


Landing Page



Executive Dashboard

Architecture diagram of the proposed solution



Technologies to be used in the solution

Backend

FastAPI
Python 3.12
Pydantic

Machine Learning

Scikit-learn
XGBoost
LightGBM
CatBoost
SHAP

Frontend

Next.js 15
TypeScript
Tailwind CSS
Shadcn UI
React Query
Recharts

AI & Data

Gemini 3.5 Flash
text-embedding-004
LangGraph
Firestore

Deployment

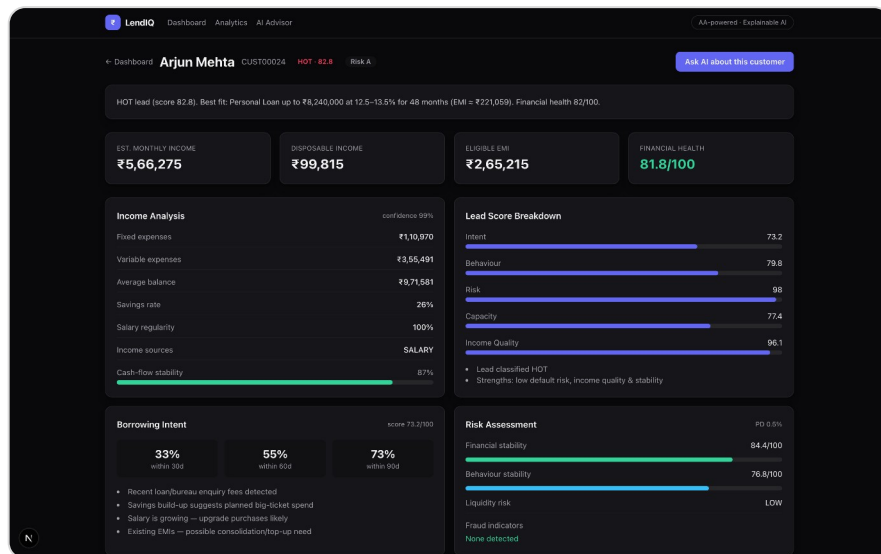
Docker
Vercel
Render

Estimated implementation cost (optional)

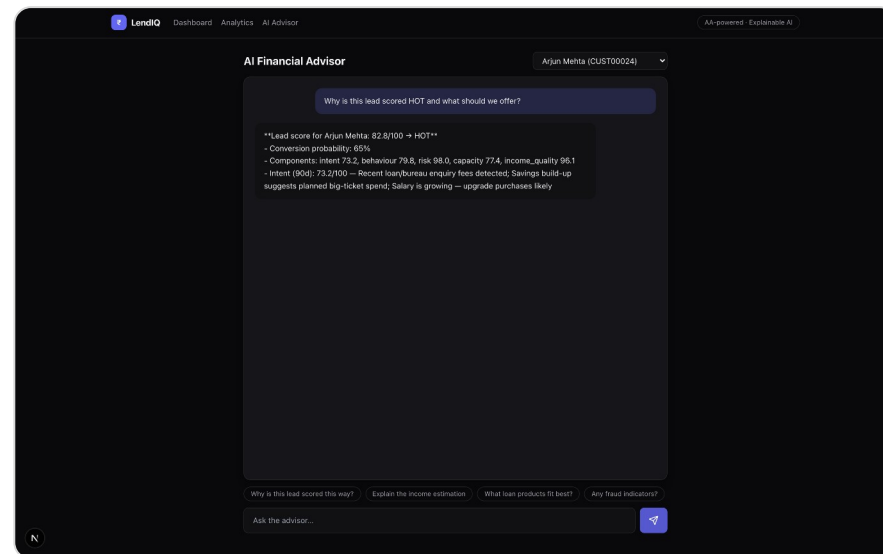
Component	Tier / Notes	Est. Monthly Cost
Backend hosting (Render)	Starter instance	₹0 – ₹1,800
Frontend hosting (Vercel)	Hobby / Pro plan	₹0 – ₹1,700
Firestore (database)	Pay-as-you-go, pilot scale	₹0 – ₹4,000
Gemini API (LLM advisor)	Pay-per-token, usage-based	₹500 – ₹4,000
Domain + misc.	Annual, amortised	≈ ₹80 / month

Pilot total: ≈ ₹2,000 – ₹10,000/month for a few hundred scored customers/month; scales with AA call volume and LLM usage. No upfront infrastructure spend.

Snapshots of the prototype



Customer Profile — Income, Lead Score, Risk



AI Financial Advisor (Gemini-grounded chat)

Prototype Performance report/Benchmarking

Model evaluation (held-out test set, 100 synthetic customers)

7.0% MAPE

Income Model (LightGBM) — RMSE ₹19,467
· MAE ₹8,751

0.62 AUC

Intent Model (XGBoost) — Precision 0.63 ·
F1 0.57

0.36 KS

Risk Model (LightGBM) — ROC-AUC 0.63 ·
Gini 0.27

Demo run — 80 scored customers

19 / 26 / 35

HOT / WARM / COLD leads

30.2%

Predicted conversion rate (target
>30%)

₹1,06,859

Average estimated monthly
income

₹40,22,625

Average loan eligibility

Additional Details/Future Development (if any)

- **Real AA integration** — connect to Setu / Sahamati AA sandbox, retrain models on real consented data
- **Auth & RBAC** — Firebase Auth with underwriter / RM role-based dashboard access
- **Model monitoring** — scheduled drift detection (already scaffolded) with automatic retraining triggers
- **Multi-lingual advisor** — Hindi & regional-language GenAI conversations, WhatsApp/SMS lead nudges
- **Model registry** — Vertex AI registry with champion/challenger traffic-split rollout
- **Core banking integration** — push HOT leads directly into the LOS/CRM for one-click underwriting

Provide links to your:

- **GitHub Public Repository:** [Add GitHub link]
- **Demo Video Link (3 Minutes):** [Add demo video link]
- **Final Product Link:** [Add live deployment link]

Summary & Business Impact

LendIQ turns raw Account Aggregator statements into pre-qualified, explainable loan leads — validated end-to-end on a working prototype.

30.2%

Predicted conversion rate in demo — target >30% achieved

7.0%

Income estimation MAPE — accurate, confidence-scored

Minutes, not days

Faster loan approval via automated scoring

Explainable by design

SHAP + GenAI advisor replace manual verification



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Thank you!